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Transistor short circuit protection

Abstract

A short circuit detection circuit includes a current terminal, a sense resistor, an amplifier, and a resistor-capacitor ladder. The sense resistor is coupled to the current terminal, and is configured to develop a sense voltage proportional to a current through the current terminal. The amplifier is coupled to the sense resistor, and is configured to generate a scaled current proportional to the sense voltage. The resistor-capacitor ladder is coupled to the amplifier, and is configured to generate a measurement voltage that represents a surface temperature rise due to the current through the current terminal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/348,521, filed Jun. 15, 2021, the entirety of which is incorporated herein by reference.

BACKGROUND

(1) Transistors, such as metal oxide semiconductor (MOS) transistors, are used as power switches in various applications. In some applications, transistors are stacked in series between two power supply rails. This arrangement is referred to as half-bridge configuration. If a short circuit develops between the two power rails through the stacked transistors, one or both of the transistors may be damaged.

SUMMARY

(2) A short-circuit detection circuit that identifies a short-circuit condition based on estimated die temperature is described herein. In one example, a short circuit detection circuit includes a current terminal, a sense resistor, an amplifier, and a resistor-capacitor ladder circuit. The sense resistor includes a first terminal coupled to the current terminal, and a second terminal. The amplifier includes a first input coupled to the first terminal of the sense resistor, a second input coupled to the second terminal of the sense resistor, and an output. The resistor-capacitor ladder circuit includes an input coupled to the output of the amplifier.

(3) In another example, a short circuit detection circuit includes a current terminal, a sense resistor, an amplifier, and a resistor-capacitor ladder. The sense resistor is coupled to the current terminal, and is configured to develop a sense voltage proportional to a current through the current terminal. The amplifier is coupled to the sense resistor, and is configured to generate a scaled current proportional to the sense voltage. The resistor-capacitor ladder is coupled to the amplifier, and is configured to generate a measurement voltage that represents a surface temperature rise due to the current through the current terminal.

(4) In a further example, a circuit includes a switching transistor, a sense transistor, a sense resistor, an amplifier, and a resistor-capacitor ladder circuit. The switching transistor includes a current terminal, and a control terminal. The sense transistor includes a first current terminal, a second current terminal, and a control terminal. The first current terminal is coupled to the current terminal of the switching transistor. The control terminal is coupled to the control terminal of the switching transistor. The sense resistor includes a first terminal and a second terminal. The first terminal of the sense resistor is coupled to the second current terminal of the sense transistor. The amplifier includes a first input, a second input, and an output. The first input is coupled to the first terminal of the sense resistor. The second input is coupled to the second input of the sense resistor. The resistor-capacitor ladder circuit includes an input coupled to the output of the amplifier.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram for an example short circuit detection circuit that detects a short-circuit condition based on estimation of surface temperature.

(2) FIG. 2 is a schematic diagram for an example resistor-capacitor (R-C) ladder circuit that models a surface portion of switching transistor semiconductor material;

(3) FIG. 3 is a schematic diagram for an example amplifier and current sources of the short circuit detection circuit of FIG. 1.

DETAILED DESCRIPTION

(4) Protecting low resistance transistors against shorts has become increasingly difficult. Short circuit currents may be high (e.g., in the range of 500 amperes) and to prevent damage transistors should be protected across all slew rates. Overcurrent detection circuitry may be provided to monitor current flow through transistors, and trigger protection when an overcurrent condition is detected. Conventional overcurrent detection circuitry measures current flow through a transistor, and compares the measured current to a threshold to identify an overcurrent condition. Because of transients generated during transistor switching, conventional overcurrent detection may be blanked out (ignored) until switching is complete and the transistor output has settled. After the transistor output has settled, if the measured current exceeds the threshold, an overcurrent condition is deemed to exist. However, the time (blanking time) during which overcurrent detection is disabled may be too long or too short given process and slew rate variation. If the blanking time is too long, and an overcurrent or short circuit condition exists, the transistor may be damaged. If the blanking time is too short, false detection of an overcurrent condition may occur.

(5) The short circuit detection circuit described herein detects a short circuit condition during transistor switching (during blanking time) by estimating the surface temperature of the switching transistor. The surface temperature of the transistor increases with current flow (energy dissipation across the transistor), and if the estimated surface temperature exceeds a threshold value, then a short circuit condition is deemed to exist. The short circuit detection circuit measures current flow through a switching transistor via a sense transistor. A high bandwidth amplifier converts the current flowing through the sense transistor to a scaled current in a low voltage domain. The scaled current is applied to a resistor-capacitor (R-C) ladder circuit that models a surface portion of the

semiconductor material of the switching transistor. When the voltage across the R-C ladder circuit exceeds a threshold, indicating a die surface temperature greater than a predetermined temperature, a short circuit condition is deemed to exist, and the switching transistor may be turned off to prevent damage.

(6) FIG. 1 is a block diagram for an example circuit **100** that includes short circuit detection based on estimated surface temperature of a switching transistor. The circuit **100** includes a short circuit detection circuit **102**, a high-side transistor **104**, and a low-side transistor **106**. The high-side transistor **104** and the low-side transistor **106** are connected in a half-bridge configuration. The high-side transistor **104** and the low-side transistor **106** may be switching transistors of a DC-DC converter, an inverter, a power factor correction circuit, or any other circuit that implements stacked transistors.

(7) The high-side transistor **104** and the low-side transistor **106** are illustrated as n-type field effect transistors. The high-side transistor **104** may be a p-type field effect transistor in some implementations of the circuit **100**. The high-side transistor **104** and the low-side transistor **106** may be gallium nitride (GaN) high electron mobility transistors (HEMTs). In FIG. 1, a drain of the high-side transistor **104** is coupled to a power supply terminal, and source of the high-side transistor **104** is coupled to a drain of the low-side transistor **106**. A source of the low-side transistor **106** is coupled to ground. The gate of the high-side transistor **104** and the gate of the low-side transistor **106** are coupled to a control circuit or a control terminal (not shown) that provides a high-side control signal to the high-side transistor **104** and the low-side control signal for turning the high-side transistor **104** and the low-side transistor **106** on or off.

(8) The short circuit detection circuit **102** is coupled to a switching node (a current node **107**) formed at the connection of the high-side transistor **104** (e.g., the source of the high-side transistor **104**) to the low-side transistor **106** (the drain of the low-side transistor **106**). The short circuit detection circuit **102** estimates the temperature of the surface semiconductor material of the low-side transistor **106** to determine whether a short circuit exists at the current node **107**. The short circuit detection circuit **102** includes a sense transistor **108**, a sense resistor **110**, a resistor **112**, an amplifier **114**, a current source **118**, a current source **120**, an R-C ladder circuit **122**, and a comparator **124**. The sense transistor **108** is a scaled-down replica of the low-side transistor **106**, and passes a sense current that is a scaled-down replica (a predetermined fraction) of the current flowing in the current node **107** and the low-side transistor **106**. The drain **108D** (a current terminal) of the sense transistor **108** is coupled to the current node **107**, and the gate **108G** (a control terminal) of the sense transistor **108** is coupled to the gate (a control terminal) of the low-side transistor **106**.

(9) The sense transistor **108** is coupled to the sense resistor **110**, and the sense current flowing through the sense transistor **108** also flows in the sense resistor **110** to develop a sense voltage across the sense resistor **110**. A terminal **110A** of the sense resistor **110** is coupled to the source terminal **108S** (a current terminal) of the sense transistor **108**, and terminal **110B** of the sense resistor **110** is coupled to ground.

(10) The sense resistor **110** is coupled to the amplifier **114**. The amplifier **114** scales the voltage across the amplifier **114** to generate a scaled current that is proportional to the sense voltage. A first input of the amplifier **114** is coupled to the terminal **110A** of the sense resistor **110**, and a second input **114B** of the amplifier **114** is coupled to ground via the resistor **112**. A terminal **112B** of the resistor **112** is coupled to ground, and a terminal **112A** of the resistor **112** is coupled to the input **114B** of the amplifier **114**.

(11) The output signal of the amplifier **114** controls the current source **118** and the current source **120**. The output **114C** of the amplifier **114** is coupled to the control input **118B** of the current source **118** and the control input **120B** of the current source **120**. The current source **118** produces, based on the output signal of the amplifier **114**, a scaled current (a feedback current) that is fed back to the input **114B** of the amplifier **114**. The output **118C** of the current source **118** is coupled

to the input **114B** of the amplifier **114**.

(12) The current source **120** produces, based on the output signal of the amplifier **114**, a scaled current that provided to the R-C ladder circuit **122**. While the high-side transistor **104**, the low-side transistor **106**, and the sense transistor **108** may operate in a high voltage domain (e.g., hundreds of volts), the amplifier **114**, the current sources **118** and **120**, the R-C ladder circuit **122**, and the comparator **124** operate in a low voltage domain (e.g., 5 volts or less). Accordingly, the output currents of the current source **118** and the current source **120** are provided in the low voltage domain. The power input **118A** of the current source **118** and the power input **120A** of the current source **120** are coupled a power supply terminal **116** (a low voltage power supply terminal).

(13) The comparator **124** is coupled to the R-C ladder circuit **122** and compares the voltage dropped across the R-C ladder circuit **122** to a threshold voltage. The voltage across the R-C ladder circuit **122** represents an estimate of the surface temperature of the semiconductor material of the low-side transistor **106**. Thus, the voltage across the R-C ladder circuit **122** increases with the current flowing in the sense transistor **108** as the surface temperature of the semiconductor material of the low-side transistor **106** increases with the current flowing in the sense transistor **108**. An input **124A** of the comparator **124** is coupled to the output **120C** of the current source **120** and the input **122A** of the R-C ladder circuit **122** for receipt of an estimated temperature signal (a measurement voltage) developed across the R-C ladder circuit **122**. An input **124B** of the comparator **124** is coupled to a reference voltage source **126** for receipt of a reference voltage representing a short circuit temperature threshold. The reference voltage may represent a surface temperature of approximately 200° Celsius in some implementations of the short circuit detection circuit **102**. An output **124C** of the comparator **124** provides a signal indicating detection of a short circuit based on estimated die surface temperature.

(14) FIG. 2 is a schematic diagram for an example R-C ladder circuit **122** that models a surface portion of the semiconductor material of the low-side transistor **106**. The R-C ladder circuit **122** includes capacitors that form the rungs of the R-C ladder circuit **122**, and resistors that are connected in series form the rail of the R-C ladder circuit **122**. The resistors emulate the thermal conductivity of the semiconductor material, and the capacitors emulate the thermal capacitance of the semiconductor material. The implementation of the R-C ladder circuit **122** shown in FIG. 2 includes ten rungs. The ten rungs include capacitors **222**, **224**, **226**, **228**, **230**, **232**, **234**, **236**, **238**, and **240**. Each capacitor may have a capacitance of about 3.2 picofarads in some implementations of the R-C ladder circuit **122**. The rail of the R-C ladder circuit **122** includes resistors **202**, **204**, **206**, **208**, **210**, **212**, **214**, **216**, **218**, **220**, and **221**. The resistor **202** may have a resistance of about 2.6 kilohms, and the resistors **204-220** may each have a value of about 5.2 kilohms in some implementations of the R-C ladder circuit **122**. The resistor **202** at one end of the rail is coupled to the input **122A**, and the resistor **221** at the other end of the rail is coupled to the input **122B**. The input **122B** is coupled to ground.

(15) A bottom plate terminal of each capacitor is coupled to the input **122B** of the R-C ladder circuit **122**. A top plate terminal of each capacitor is coupled to at least one of the resistors of the rail. For example, the top plate terminal of the capacitor **222** is coupled to the resistor **202** and the resistor **204**, the top plate terminal of the capacitor **224** is coupled to the resistor **204** and the resistor **206**, etc.

(16) In the R-C ladder circuit **122**, each rung emulates the thermal capacitance of approximately one micron of depth of the surface semiconductor material of the low-side transistor **106**. Thus, given the ten rungs shown in FIG. 2, the R-C ladder circuit **122** emulates a 10-micron depth of semiconductor material. In short circuit conditions of 200 nanoseconds or less, a heatwave in GaN does not diffuse more than 10 microns. Some implementations of the R-C ladder circuit **122** may include a different number of rungs to represent a different depth of semiconductor material.

(17) FIG. 3 is a schematic diagram for examples of the amplifier **114** and current sources **118** and **120** of the short circuit detection circuit **102**. The amplifier **114** includes a differential input pair

301 and a bias circuit **305**. The differential input pair **301** has common-mode gate configuration, and includes the transistor **302** and the transistor **304**. The output **114C** of the amplifier **114** is coupled to the drain of the transistor **304**. The source of the transistor **304** is coupled to the input **114B**. The source of the transistor **302** is coupled to the input **114A** via the resistor **313**.

(18) The bias circuit **305** sources current to the differential input pair **301**, and includes the transistor **306**, the transistor **308**, the transistor **310**, and the transistor **312**. The source of the transistor **310** and the source of the transistor **312** are coupled to the power supply terminal **116**. The drain of the transistor **310** is coupled to the source of the transistor **306**, and the drain of the transistor **312** is coupled to the source of the transistor **308**. The drain of the transistor **306** is coupled to the drain of the transistor **302**, and the drain of the transistor **308** is coupled to the drain of the transistor **304**.

(19) The current source **118** includes a transistor **314**, a transistor **316**, and a resistor **318**. The source of the transistor **314** is coupled to the power supply terminal **116**. The gate of the transistor **314** is pulled down such that the transistor **314** is always on. The drain of the transistor **314** is coupled to the source of the transistor **316** via the resistor **318**. The gate of the transistor **316** is coupled to the output **114C** of the amplifier **114**, so that current flow through the transistor **316** is controlled by the output signal of the amplifier **114**. The drain of the transistor **316** is coupled to the input **114B** of the amplifier **114**.

(20) The current source **120** includes selectable current sources **319**, **321**, and **323**. Some implementations of the current source **120** may include a different number of selectable current sources (more or less than three selectable current sources). The current sources **319**, **321**, and **323** may be individually enabled or disabled via a current selection value. The current selection value may be set at manufacture based on the current rating of the low-side transistor **106**. The current source **319** includes a transistor **320**, a transistor **322**, and a resistor **324**. The source of the transistor **320** is coupled to the power supply terminal **116**. The gate of the transistor **320** is controlled by the current selection value to enable or disable the current source **319**. The drain of the transistor **320** is coupled to the source of the transistor **322** via the resistor **324**. The gate of the transistor **322** is coupled to the output **114C** of the amplifier **114**, so that current flow through the transistor **322** is controlled by the output signal of the amplifier **114**. The drain of the transistor **322** is coupled to the output **120C** for driving the R-C ladder circuit **122**.

(21) The current source **321** includes a transistor **326**, a transistor **328**, and a resistor **330**. The source of the transistor **326** is coupled to the power supply terminal **116**. The gate of the transistor **326** is controlled by the current selection value to enable or disable the current source **321**. The drain of the transistor **326** is coupled to the source of the transistor **328** via the resistor **330**. The gate of the transistor **328** is coupled to the output **114C** of the amplifier **114**, so that current flow through the transistor **328** is controlled by the output signal of the amplifier **114**. The drain of the transistor **328** is coupled to the output **120C** for driving the R-C ladder circuit **122**.

(22) The current source **323** includes a transistor **332**, a transistor **334**, and a resistor **336**. The source of the transistor **332** is coupled to the power supply terminal **116**. The gate of the transistor **332** is controlled by the current selection value to enable or disable the current source **323**. The drain of the transistor **332** is coupled to the source of the transistor **334** via the resistor **336**. The gate of the transistor **334** is coupled to the output **114C** of the amplifier **114**, so that current flow through the transistor **334** is controlled by the output signal of the amplifier **114**. The drain of the transistor **334** is coupled to the output **120C** for driving the R-C ladder circuit **122**.

(23) In FIG. 3, the R-C ladder circuit **122** also includes a reset input **122C** for receiving a reset signal. The reset signal, when active, discharges the capacitors of the R-C ladder circuit **122** in preparation for a next temperature estimation. The reset signal activates the transistor **346** to pull the input **122A** to ground to discharge the capacitors of the R-C ladder circuit **122**. The reset signal provided at the reset input **122C** may close switches coupled in parallel with one or more of the capacitors of the R-C ladder circuit **122** to discharge the capacitors.

(24) Short circuit detection as described herein with respect to the low-side transistor **106** may also be applied to the high-side transistor **104** by coupling an instance of the short circuit detection circuit **102** across the high-side transistor **104**. For example, the drain **108D** of the sense transistor **108** may be coupled to the drain of the high-side transistor **104**, the gate **108G** of the sense transistor may be coupled to the gate of the high-side transistor **104**, and the terminal **110B** of the sense resistor **110** may be coupled to the source of the high-side transistor **104**.

(25) In this description, the term “couple” may cover connections, communications or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is directly coupled to device B; or (b) in a second example, device A is indirectly coupled to device B through intervening component C if intervening component C does not substantially alter the functional relationship between device A and device B, so device B is controlled by device A via the control signal generated by device A.

(26) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A circuit comprising: a first input terminal; a second input terminal; a transistor having a control terminal coupled to the second input terminal, a first current terminal coupled to the first input terminal, and a second current terminal; a first amplifier having a first input terminal coupled to the second current terminal of the transistor, and an output; a first current source having a control terminal coupled to the output of the first amplifier, and an output terminal; and a second amplifier having a first input terminal coupled to the output terminal of the first current source, a second input terminal, and an output terminal.
2. The circuit of claim 1, wherein: the transistor is a first transistor; the first input terminal is adapted to be coupled to a current terminal of a second transistor; the second input terminal is adapted to be coupled to a control terminal of a third transistor; and the current terminal of the second transistor is coupled to a current terminal of the third transistor.
3. The circuit of claim 2, wherein: the second input terminal of the second amplifier is coupled to a voltage reference.
4. The circuit of claim 3, wherein: the voltage reference is capable of corresponding to a temperature of the second transistor.
5. The circuit of claim 1, wherein: the second current terminal of the transistor is coupled to a first terminal of a resistor.
6. The circuit of claim 1, wherein: the first amplifier has a second input terminal; the second input terminal of the first amplifier is coupled to a first terminal of a resistor; and a second terminal of the resistor is coupled to a ground terminal.
7. The circuit of claim 6, wherein: the output terminal of the first current source is coupled to the second input terminal of the first amplifier.
8. The circuit of claim 1, further comprising: a second current source having a control terminal coupled to the output of the first amplifier.
9. The circuit of claim 1, wherein: the output terminal of the second amplifier is coupled to a resistor-capacitor ladder circuit.
10. The circuit of claim 9, wherein the resistor-capacitor ladder circuit includes: a first capacitor having a first terminal and a second terminal; a first resistor having a first terminal and a second terminal coupled to the first terminal of the first capacitor; a second capacitor having a first terminal and a second terminal; and a second resistor having a first terminal coupled to the first terminal of the first capacitor and the second terminal of the first resistor, a second terminal of the second resistor coupled to the first terminal of the second capacitor.

11. A circuit comprising: a transistor; a first amplifier having an output, the first amplifier coupled to the transistor; a plurality of current sources coupled to an output of the first amplifier; a resistor-capacitor ladder circuit coupled to a first current source of the plurality of current sources; a voltage reference; and a second amplifier coupled to the voltage reference and the resistor-capacitor ladder circuit, wherein the second amplifier is capable of outputting a signal indicating detection of a short circuit.
 12. The circuit of claim 11, wherein the resistor-capacitor ladder circuit includes: a plurality of rungs, wherein each of the plurality of rungs includes a capacitor.
 13. The circuit of claim 12, wherein: each of the plurality of rungs includes a resistor.
 14. The circuit of claim 11, wherein: the transistor is coupled to a resistor.
 15. The circuit of claim 14, wherein: the first amplifier is capable of receiving a voltage across the resistor; and the voltage is based on a current through the transistor.
 16. The circuit of claim 15, wherein: the transistor is a first transistor; the current is a first current; the first current is proportional to a second current in a second transistor; and the voltage is based on the first current through the first transistor.
 17. A method: receiving, by a first transistor, a first current; conducting, by the first transistor, a second current, wherein the second current is proportional to the first current, and wherein the second current is conducted through a resistor to develop a first voltage; generating, by an amplifier, a first scaled current proportional to the first voltage; outputting, by the amplifier, the first scaled current to a plurality of current sources; outputting, by a first one of the plurality of current sources, a second scaled current; receiving, by a resistor-capacitor ladder circuit, the second scaled current; and outputting, by a comparator, a signal based on a comparison of a voltage from the resistor-capacitor ladder circuit and a reference voltage.
 18. The method of claim 17, further comprising: outputting, by a second one of the plurality of current sources, a feedback current; and receiving, by the amplifier, the feedback current.
 19. The method of claim 17, wherein: the signal is capable of indicating a determination of a short circuit.
 20. The method of claim 17, further comprising: outputting, by a second one of the plurality of current sources, a feedback current to the amplifier.
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